

Saqib Shahab

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PROFESSIONAL SUMMARY

Software Engineering student with hands-on experience building AI/ML pipelines and full-stack web applications. Fine-tuned a DONUT Vision-Language Model to a 0.19 training loss for automated document data extraction, and shipped production React.js components for an HR management platform during a front-end internship. Comfortable working across the stack: model training and evaluation (PyTorch, TensorFlow), backend APIs (Flask, Node.js), and frontend delivery (React.js), with practical experience in Agile team environments.

EDUCATION

Bachelor of Science in Software Engineering — University of Engineering and Technology (UET), Taxila

Sep 2022 – Jun 2026

PROFESSIONAL EXPERIENCE

AI Trainee — NASTP, Rawalpindi

May 2025 – Jun 2026

- Fine-tuned the DONUT Vision-Language Model in PyTorch to automate JSON extraction from complex document images, reducing training loss to 0.19.
- Built a custom image-to-text preprocessing pipeline that improved document-understanding accuracy on a limited labeled dataset, cutting the manual data-cleaning time needed before model training.

Front-End Developer Intern — OraDigitals, Okara

Jun 2025 – Aug 2025

- Architected a responsive HR Management System (OIMS) in React.js, restructuring component state management to reduce unnecessary re-renders and speed up dashboard load times.
- Built 10+ reusable React.js components adopted across multiple product modules, cutting duplicate code and shortening delivery time for new features built during Agile sprints.

PROJECTS

DonutDoc — Final Year Project: AI Document Processing System

- Built a CUDA-accelerated Python/FastAPI backend on Google Colab using the DONUT Vision-Encoder-Decoder model for OCR-free data extraction from document images.
- Developed a Node.js/Express API gateway to manage asynchronous request queuing, secure connections to the model backend, and persistence of extracted JSON data in MongoDB.
- Built a React.js frontend supporting batch document uploads with real-time extraction results, giving users an interactive review workspace.

Apex Auto Automation Agent — Autonomous Task Automation Engine

- Built a Python automation agent using Selenium and the Grok API to execute multi-step digital workflows (including booking processes), removing the need for manual, repetitive input.
- Implemented a modular framework with secure credential management, automated error handling, and logging to support reliable, unattended runs.

AMS PRO — AI-Powered Full-Stack Attendance System

- Built a full-stack attendance system (React.js, Flask, MongoDB) with real-time OpenCV facial recognition to automate attendance logging.
- Deployed the system on Render with a responsive, cloud-native architecture accessible across devices.

ResumAgent — RAG-Based Document Analysis Assistant

- Built a Retrieval-Augmented Generation (RAG) system with LangChain and FAISS to answer questions grounded in academic and professional documents.
- Designed a vector-based retrieval pipeline to surface relevant document context and improve the relevance of generated answers.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, C#, HTML/CSS

Frameworks & Libraries: PyTorch, TensorFlow, React.js, Flask, Node.js, OpenCV, LangChain, FAISS

Tools & Platforms: Git/GitHub, VS Code, Jupyter, Kaggle, MATLAB, Render, Google Colab

Databases: MongoDB, MySQL, PostgreSQL, Firebase